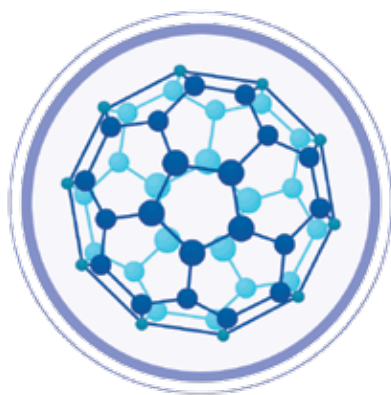




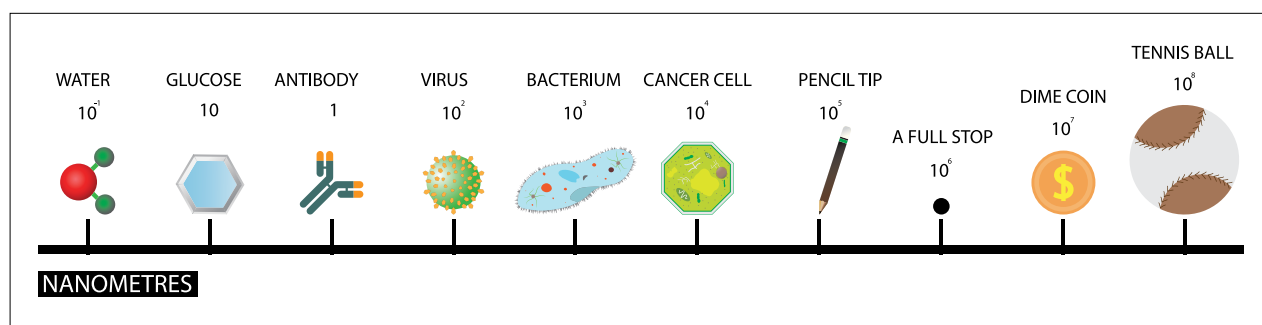
NANOSCALE

Every single Digit reader will immediately point to their phone or PC when thinking about nanotech that they are already using. If you have a PC or phone made in the last two years, you're already using 10 nano-

adult or teen, are measured on a metre scale, and range between 1.3 to 2 metres in height. The width of one of your fingers is 100 times smaller at about 1 to 2 cm. Go ten times smaller, and you arrive at the width/diameter of one grain of beach sand. Ten times smaller again and you arrive at the thickness of a rather "fat" sample of a human hair (average is 0.08 mm, or 80 μm , or 80 microns). 100 microns is also the approximate thickness of the inside pages of our print magazine. Below this size, our human eyes become useless, and we need the help of microscopes to see anything smaller. A red blood cell is about 7 μm , or very roughly, about 10 times thinner than a human hair. At around the 1 μm mark we find

10 times smaller than a virus, DNA is 2 nm in width, and a hydrogen atom is 0.1 nm (or 1 angstrom) in diameter. When we're talking about nanotechnology, we're talking about pretty much man-made that is between 1 nm and 100 nm in size.

What makes nanotechnology so important is the fact that in the world of the really small, it's not classical physics that we are used to that plays out, but rather, the quantum physics that none of us can truly understand, and which the world's smartest people are still struggling to explain. At a nano-scale, materials behave very differently than they do on the macro scale of a human, and this brings interesting properties and possibilities that we are exploring, and thus



Everything between the size of an antibody and a virus is considered nanotech

metre (nm) or smaller technology. When Digit started in 2001, the cutting edge of semiconductor fabrication was 130 nm, and a mere 19 years later we're sitting at 5 nm fabs already being used... that's 26 times smaller... in just 19 years!

A crash course in the world of nano-sizes and quantum effects follows. A nanometre is one billionth of a metre (10^{-9}), which is really hard to comprehend unless we put things into perspective. You, as a human

bacteria, though we should add that there is a huge range of sizes that bacteria come in. E-coli, the bacteria most of us know about because it lives inside our digestive system, is about 2 μm in size. The world of viruses is about 10 times smaller than bacteria, with the current SARS-CoV2 virus being measured to be between 60 and 140 nm in size – let's take an average of 100 nm, or about 0.1 μm , 0.0001 mm, or 0.0000001 metres in size. A 10 nm silicon chip is

the whole scientific discipline of nanoscience was birthed. Nanotechnology is the reason why modern technology means that you were using more computing power by doing one single Google search (way back in 2012!), than all of the computing power used by NASA in the Apollo missions in the 1960s – including both ground and spacecraft computers, across the entire 17 missions that spanned 11 years of computations! Think about that the next time you

Combating COVID-19

An overview of the various nanotechnology based approaches that can be used to improve the drug delivery in patients suffering from COVID-19.
<http://dgit.in/nanocov>

Star Trek

This scientific paper explores both the benign and evil use of nanotechnology in the Star Trek universe, both by the future humans and the borg.
<http://dgit.in/nanotrek>

Cosmetics

The cosmetics industry is actually at the forefront of bringing nanotechnologies to the consumer, with 13 percent of all current nanotech products in the market being cosmetics.
<http://dgit.in/nanocosm>